Adaptive Filters

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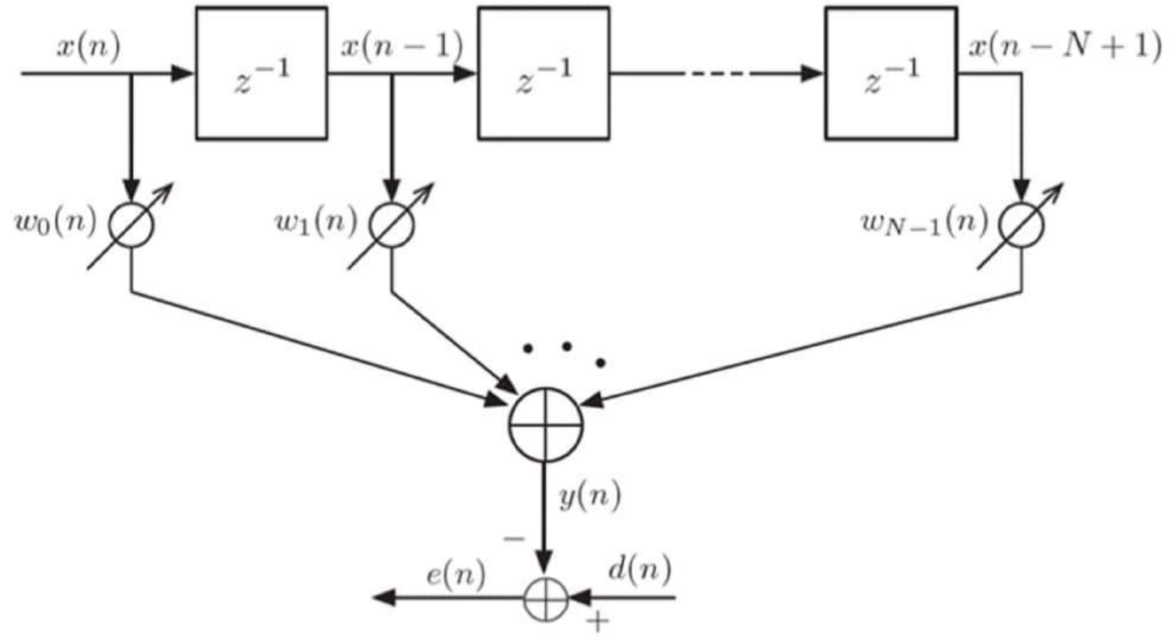
Adaptive Filters: Applications

- Noise Cancellation (Active Noise Control)
- Echo Cancellation
- System Identification / Modelling
- Channel Equalization in Communication systems
- Adaptive Beam Forming in Radar, Sonar, Communication Systems
- Adaptive Filtering in Biomedical Signal Processing (ECG, EEG, EMG etc)
- Prediction & Forecasting (Financial, Weather, Speech etc)

Adaptive Filter: Least Mean Squares (LMS)



Summary of Equations:

- **Filter Output:** $y(n) = \sum_{i=0}^{N-1} w_i(n) \cdot x(n-i)$
- **Error Signal:** $e(n) = d(n) - y(n)$
- **Coefficient Update:** $w_i(n+1) = w_i(n) + 2\mu \cdot e(n) \cdot x(n-i)$